

DATA SCIENCE CAPSTONE PROJECT REPORT

SpaceX Falcon 9 First-Stage Landing Prediction

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Submission date: 13 September 2026

GitHub repository: <https://github.com/kunalji-jj/data-science-capstone-kunal-singh>

Project focus: collecting, cleaning, exploring, visualising and modelling SpaceX launch data to understand and predict first-stage landing success.

Academic submission report

Important Submission Note

This report is structured to cover every criterion identified in the grading feedback. Before final submission, the actual completed notebook outputs, screenshots, model metrics, SQL result tables, Folium map and Plotly Dash dashboard must be inserted in the marked evidence areas. The report must not be submitted with example values or empty evidence boxes.

The written explanations below describe the project workflow and the evidence that should be shown. Numerical model results are intentionally not fabricated because they must match the learner's executed notebooks.

1. Executive Summary

This capstone project investigates whether the first stage of a SpaceX Falcon 9 launch successfully lands after the mission. Reusing the first stage can reduce launch costs, so predicting landing success is an important operational and business question. The project uses publicly available SpaceX launch information collected through an API and transformed into an analysis-ready dataset.

The analysis combines data collection, data wrangling, exploratory data analysis, SQL queries, interactive visualisation, geospatial analysis with Folium, an interactive Plotly Dash dashboard, and supervised machine-learning classification. The target variable is the first-stage landing outcome, represented as a binary success/failure label.

The final conclusions must be based on the actual executed notebook outputs. The completed submission should identify the strongest relationships between launch-site, payload, orbit, booster/version and landing outcome, compare classification models, and explain the practical limitations of predicting landing success from historical launch data.

Executive summary evidence: final findings chart or results table

Insert the actual screenshot/output from the completed IBM notebook.

2. Introduction

SpaceX's Falcon 9 launch system is designed to reuse its first stage. Successful recovery can improve cost efficiency, but landing is technically difficult and depends on mission and vehicle characteristics. A data-science approach can help identify patterns in historical launches and estimate the likelihood of a successful landing.

Project objective

To collect and analyse SpaceX launch data and build a classification model that predicts whether the Falcon 9 first stage will land successfully.

Research questions

- How can SpaceX launch data be collected and transformed into a reliable analysis dataset?
- Which launch-site, orbit, payload and booster-related features are associated with landing success?
- What patterns are visible through charts, SQL analysis and an interactive map?
- Which classification algorithm provides the most useful predictive performance?
- What practical insights and limitations should be considered when interpreting the results?

Author and date: Kunal Singh — 13 September 2026

3. Data Collection Methodology

The project uses the SpaceX REST API and supporting public launch information. The collection process retrieves launch records, rocket/booster information, payload details, launch-site information and landing-related attributes. The raw responses are converted into tabular form for analysis.

Collection workflow

- Send requests to the SpaceX API endpoint used in the IBM lab.
- Inspect the JSON response and identify relevant nested fields.
- Extract launch dates, launch-site names, orbit, payload mass, booster/version and landing outcome.
- Store the extracted records in a pandas DataFrame.
- Save the cleaned dataset so that later notebooks use a consistent source.

Actual API request and first DataFrame output

Insert the actual screenshot/output from the completed IBM notebook.

Evidence required: screenshot showing the executed request, response inspection and a real DataFrame preview.

4. Data Wrangling Methodology

Data wrangling prepares the raw launch records for exploratory analysis and modelling. The process includes selecting relevant columns, flattening nested JSON fields, handling missing values, converting data types, creating a binary target variable and checking the final dataset for consistency.

Key wrangling operations

- Select the launch-site, orbit, payload, booster, date and landing-outcome fields.
- Convert nested API structures into flat columns.
- Convert dates and numeric fields to appropriate data types.
- Handle missing payload masses and unavailable categorical values using a documented rule.
- Encode landing success as a binary target, for example 1 for success and 0 for failure.
- Use one-hot encoding for categorical predictors where required.
- Check duplicate rows, null values, class balance and final dimensions.

Actual cleaning code and before/after DataFrame output

Insert the actual screenshot/output from the completed IBM notebook.

The final report should state the actual number of rows and columns after cleaning, taken directly from the executed notebook.

5. EDA and Data Visualisation Methodology

Exploratory data analysis is used to understand the structure of the dataset before modelling. The analysis examines launch frequency over time, landing outcomes by category, payload distributions, orbit patterns and relationships between categorical features and the target variable.

Visualisation principles

- Use clear titles, axis labels and legends.
- Display counts or percentages consistently.
- Explain what each chart shows and why it matters.
- Avoid presenting a chart without an interpretation.
- Compare success and failure groups where appropriate.

Actual EDA charts with captions and interpretations

Insert the actual screenshot/output from the completed IBM notebook.

6. EDA with Visualisation

Recommended visual 1 — Launches over time

A time-series or count chart should show how the number of launches changed over the observation period. The accompanying explanation should identify periods of increased activity and relate them to the development of the Falcon 9 programme.

Insert actual launch-frequency chart

Insert the actual screenshot/output from the completed IBM notebook.

Recommended visual 2 — Landing success by orbit

A grouped or stacked chart should compare landing outcomes across orbit categories. The interpretation should identify which orbit groups have higher or lower observed success rates, while noting that small groups may produce unstable percentages.

Insert actual orbit-versus-landing chart

Insert the actual screenshot/output from the completed IBM notebook.

Recommended visual 3 — Payload and landing outcome

A scatter, box or bar chart should examine payload mass in relation to landing outcome. The explanation should discuss whether payload mass appears to influence success and whether the relationship differs by orbit or booster type.

Insert actual payload-versus-outcome chart

Insert the actual screenshot/output from the completed IBM notebook.

7. EDA with SQL

SQL analysis provides a second method for exploring the cleaned launch dataset. The queries should be executed in the IBM SQL environment or the environment used in the course, and the report must show both the query and its returned result.

Query 1: View the first records

```
SELECT *  
FROM SPACEXTBL  
LIMIT 10;
```

Query 2: Count launches by launch site

```
SELECT Launch_Site, COUNT(*) AS Launch_Count  
FROM SPACEXTBL  
GROUP BY Launch_Site  
ORDER BY Launch_Count DESC;
```

Query 3: Average payload mass by booster version

```
SELECT Booster_Version,  
AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass  
FROM SPACEXTBL  
GROUP BY Booster_Version  
ORDER BY Average_Payload_Mass DESC;
```

Query 4: Successful landings by orbit

```
SELECT Orbit, SUM(Class) AS Successful_Landings,  
COUNT(*) AS Total_Launches  
FROM SPACEXTBL  
GROUP BY Orbit  
ORDER BY Successful_Landings DESC;
```

Insert screenshots of executed SQL queries and result tables

Insert the actual screenshot/output from the completed IBM notebook.

8. Folium Interactive Map

A Folium map is used to display the geographic locations of SpaceX launch sites. Each launch site is represented by a marker, and the marker pop-up should contain the site name and relevant information. The map helps communicate the geographic distribution of launch operations.

Required map evidence

- Actual interactive map rendered from the completed notebook.

- Markers for the launch sites used in the dataset.
- A visible pop-up or label identifying at least one launch site.
- A short interpretation explaining why launch-site location is relevant to the analysis.

Insert actual Folium map screenshot with visible markers

Insert the actual screenshot/output from the completed IBM notebook.

9. Plotly Dash Application

The Plotly Dash application provides an interactive interface for exploring launch records. A user should be able to select a launch site or another available category and view an updated chart or summary. The dashboard converts the analysis into an accessible decision-support view.

Dashboard components to document

- Dropdown or selection control.
- Callback that updates the displayed chart or table.
- Plotly chart showing launch success/failure or another relevant measure.
- Clear title, labels and user instructions.

Insert actual running Plotly Dash screenshot showing control and chart

Insert the actual screenshot/output from the completed IBM notebook.

Repository file: `Plotly_Dash/app.py`

10. Predictive Analysis — Classification

The predictive task is binary classification: estimate whether a Falcon 9 first stage will land successfully. The cleaned features are divided into training and testing sets. Categorical variables are encoded, and numerical variables are prepared consistently between training and testing data.

Models to document

- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbours

The report should explain the train/test split, preprocessing approach, random-state choice, evaluation metrics and any hyperparameter tuning performed. The same test set should be used for a fair comparison.

Evaluation metrics

- Accuracy — proportion of correctly classified observations.
- Precision — proportion of predicted successes that were actual successes.
- Recall — proportion of actual successes correctly identified.
- F1-score — balance between precision and recall.
- Confusion matrix — counts of true positives, true negatives, false positives and false negatives.

Insert actual model-training code and evaluation output

Insert the actual screenshot/output from the completed IBM notebook.

11. Predictive Analysis Results

The final results section must be completed using the actual output generated by the predictive-analysis notebook. The model comparison table below is a reporting format; its values must be copied from the executed notebook rather than estimated.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	Actual value	Actual value	Actual value	Actual value
Support Vector Machine	Actual value	Actual value	Actual value	Actual value
Decision Tree	Actual value	Actual value	Actual value	Actual value
K-Nearest Neighbours	Actual value	Actual value	Actual value	Actual value

Insert actual confusion matrix and model-comparison chart

Insert the actual screenshot/output from the completed IBM notebook.

Interpretation to complete: Identify the best-performing model using the chosen metric, explain the trade-offs, and discuss whether the result is reliable given the dataset size, class balance and feature limitations.

12. Conclusion

This project demonstrates an end-to-end data-science workflow using SpaceX launch data. The work covers data acquisition, cleaning, exploratory analysis, SQL, visual communication, geospatial mapping, dashboard development and classification modelling.

The final conclusion must be tied to the actual evidence. It should summarise the most important patterns discovered, state which model performed best on the test data, and explain how the findings could support further analysis of launch reliability and first-stage recovery.

Limitations

- The dataset contains a limited number of historical launches.
- Some variables may be missing, simplified or unevenly distributed.
- Historical patterns do not guarantee future landing outcomes.
- Model performance may be affected by class imbalance and the selected features.
- The project predicts landing success, not the complete cost or operational feasibility of a mission.

13. Creativity and Presentation Quality

Creative elements include combining static EDA with SQL, a geographic Folium map and an interactive Plotly Dash application. The report should use consistent headings, readable charts, concise captions and a clear narrative from business problem to model result.

To demonstrate creativity, include a dashboard view that allows the reader to compare launch sites or landing outcomes interactively, and use a visual callout to highlight the most important evidence-based finding.

14. Innovative Insights

The final submission should present at least one insight that goes beyond describing a chart. For example, the learner may compare success rates across launch sites while controlling for orbit or payload group, identify a category with an unexpectedly high or low success rate, or explain how a model's errors affect practical interpretation.

Insight statement format

"The analysis shows that [actual finding]. This is supported by [chart/table/model output]. The finding matters because [practical interpretation]. However, [limitation or caution]."

Insert the actual innovative insight chart/table and explanation

Insert the actual screenshot/output from the completed IBM notebook.

15. GitHub Repository and Submission Checklist

Repository URL

<https://github.com/kunalji-ji/data-science-capstone-kunal-singh>

Required item	Status before submission
README.md with project description and author	Verify
Data collection notebook	Verify actual completed file
Data wrangling notebook	Upload completed file

EDA visualisation notebook	Upload completed file
SQL analysis notebook / SQL outputs	Upload completed file
Folium map notebook / HTML	Upload completed file
Predictive analysis notebook	Upload completed file
Plotly Dash app.py	Upload and test
Dataset or reproducible data instructions	Verify
Final PDF presentation/report	Upload

Final verification checklist

- Every chart is generated from the actual project dataset.
- Every SQL query has a visible executed result.
- The Folium map is an actual rendered map with markers.
- The Plotly Dash screenshot shows a working interactive control and output.
- The predictive results contain real metrics and a confusion matrix.
- The GitHub URL opens publicly and contains the completed files.
- No text such as “insert screenshot here”, “example value” or “to be added” remains in the submitted PDF.

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